Why is the Bayesian network so confident?

Diagnosis, and a plan for a Bayesian uncertainty for `HEFTNet`
from a mixture of Laplace approximations

For approval before any code is written

October 31, 2026 (revised: mixture of Laplace approximations)

Abstract

The full-data Bayesian network (`runs/BNN_poisson_full`) gives a posterior mean almost as accurate as the fine-tuned ensemble, but a posterior standard deviation of only 0.5% of the POWHEG noise, so that its pulls have a standard deviation of 1.37 just above threshold. Before a Bayesian result appears in a paper, we must know *why*. A posterior that is too narrow can have four causes: (A) the chain has not explored the mode it sits in, (B) it has not visited the other modes, (C) the likelihood claims the network describes POWHEG better than it does, and (D) the prior. A and B are failures of the computation; C and D are choices in the model. The evidence in hand points strongly to A, and to B and C. We therefore replace the Markov chain by an approximation that suffers from neither A nor B: the posterior is approximated by a *mixture of Gaussians*, one about each of the 20 networks of the fine-tuned ensemble, each obtained by linearising the network about that point (the linearised Laplace approximation of MacKay). It needs no sampling, takes of order an hour on the 5090, and gives the within-mode and between-mode uncertainties separately. Cause C is handled, as in the appendix, by a model-error term. Nothing is written until you approve; the decisions are collected in section 9.

The message for the paper

Bayes' theorem is not on trial. The posterior answers a precisely posed question: given *this* network, *this* prior and a likelihood that says the POWHEG noise is the only discrepancy, how well are the network's parameters, and hence its predictions, determined? If the computation is done correctly, that answer is right; if the predictions are then too confident, the fault lies in the question, namely in the likelihood, which ignores the error of the model itself. The plan is designed to end in the statement: "the posterior, approximated by a mixture of Laplace approximations about the ensemble networks, together with a model-discrepancy term, gives calibrated uncertainties"; or in a clear account of what prevents it. The Markov chain results are then explained, not hidden: a single chain of this length cannot sample a 3,944-dimensional posterior with this geometry.

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1 What we observed

	median s/σ_k	s.d. of pull, 255–555 GeV	s.d. of pull, $>$ 555 GeV
Bayesian network (`BNN_poisson_full`)	0.005	1.37	1.06
fine-tuned ensemble, bootstrap	0.056	1.15	1.04
+ model error, $a(m_{hh})$	0.32	1.02	1.00

Table 1: Calibration on the 1,000 test spectra (from the draft appendix).

The earlier runs (`docs/bnn/bnn_run1.pdf` to `bnn_run3.pdf`: a Laplace likelihood; the Gaussian likelihood with known variances on 10% of the data, with fewer and then more leapfrog steps) showed the same symptom, the first of them a chain that hardly moved at all. The full-data run is the best of them, and it is the one analysed here. The run: Gaussian likelihood with the known POWHEG variances (the `poisson` option), hierarchical Gaussian prior with 16 Gibbs-sampled precisions [2], warm start from `runs/L1Loss`, 1,000 warm-up iterations, 1,000 samples, 800 leapfrog steps, acceptance rate 0.72, final step size $\varepsilon \approx 7.7 \times 10^{-4}$. The effective sample size of the two monitored predictions is about 135. The posterior mean differs from the ensemble mean by a median of 0.8 ensemble standard deviations: a perfectly sensible central value, with an implausibly narrow band around it.

2 Four ways a posterior can be too narrow

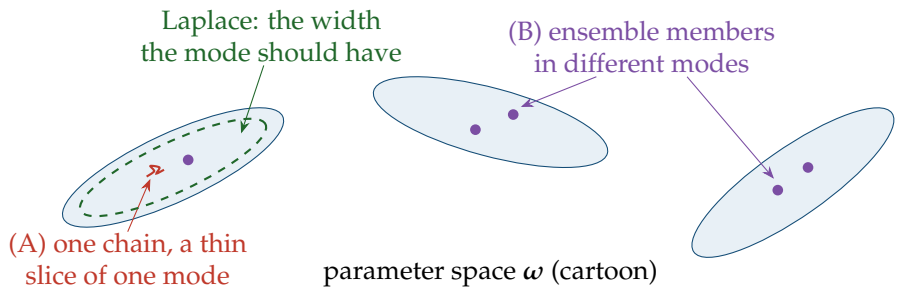


Figure 1: Cartoon of the posterior: several separated modes (blue). A chain that mixes poorly covers a thin slice of one mode (red); the linearised Laplace approximation gives the width a well-sampled mode should have (green); ensemble members sit in different modes (purple). Causes C and D change the shape of the ellipses themselves.

A. Poor sampling within a mode. HMC moves along stiff and soft directions alike with one step size, set by the stiffest. If the mass matrix does not equalise the curvatures, the soft directions are explored by a slow random walk, and a chain of 10^3 iterations sees only a fraction of their width.

B. Modes not visited. Networks have many separated modes (equally good solutions that are not related by a symmetry). A single chain stays in one [4]. The seeds ensemble shows that about 80% of the ensemble variance is of this kind.

C. An over-confident likelihood. The likelihood $\prod_k \mathcal{N}(t_k; f_k, \sigma_k^2)$ asserts that the network can represent the true cross section exactly and that the POWHEG noise is the only discrepancy. The model-error study says otherwise: on data not used in the training, a further variance $a\,\sigma_k^2$ with $a = 0.1$–0.75 is needed.

D. The prior. With 2.7×10^5 training rows the likelihood dominates every direction that affects the predictions at points like the training points; the prior mainly controls the directions the data leave free, which hardly affect the predictions. A factor of ten from the prior would be surprising, but it is cheap to check.

A and B are different questions

Even a perfect sampler need not make the posterior as wide as the ensemble. Whether the modes of fig. 1 contribute to the posterior depends on their probability *mass*: if one mode has a mass e^{100} times that of the others, the posterior is that mode, and the ensemble spread, although a fair description of the variability of the *training procedure*, is not part of the Bayesian answer. We return to this when choosing the weights of the mixture (section 4.5).

3 Evidence already in hand

3.1 An identity: the effective number of parameters

For a model linear in its parameters, $f = J\omega$, with Gaussian noise of known variances σ_k^2 and a Gaussian prior of precision A, the posterior covariance is $\Sigma = (J^\mathsf{T} W J + A)^{-1}$ with $W = \mathrm{diag}(1/\sigma_k^2)$,

and

$$\sum_{k=1}^{N} \frac{\mathrm{Var}_{\mathrm{post}}[f_k]}{\sigma_k^2} = \mathrm{tr}\left(J^{\mathsf{T}} W J \Sigma\right) = \sum_i \frac{\lambda_i}{\lambda_i + \alpha_i} \equiv \gamma, \tag{1}$$

where, for a prior $A = \alpha I$, the λ_i are the eigenvalues of $J^{\mathsf{T}} W J$. This is MacKay's "number of well-determined parameters" [1]: each direction pinned down by the data contributes one, each direction left to the prior contributes nothing, so $0 \le \gamma \le p$. Near a mode a network is approximately linear in its parameters, so eq. (1) applies approximately: the sum over the training rows of s_k^2/σ_k^2, with s_k the posterior standard deviation of the prediction, counts the well-determined parameters.

What the run says

Estimating the sum as N times the average of s_k^2/σ_k^2 over the test spectra (right panel of fig. 2), the Bayesian run gives $\gamma \approx 50$ out of $p = 3{,}944$ parameters. For a network that represents 23 functions of m_{hh} well enough to give $\chi^2/N \approx 1.1$ on 80 bins over a 5-dimensional parameter space, that is far too few. The fine-tuned ensembles, which are not posteriors, give 3,100–4,800: the order of magnitude a well-determined network of this size should have. A correctly sampled single mode is therefore expected to give a width of order the ensemble spread, roughly ten times what the chain reports. This points to cause A.

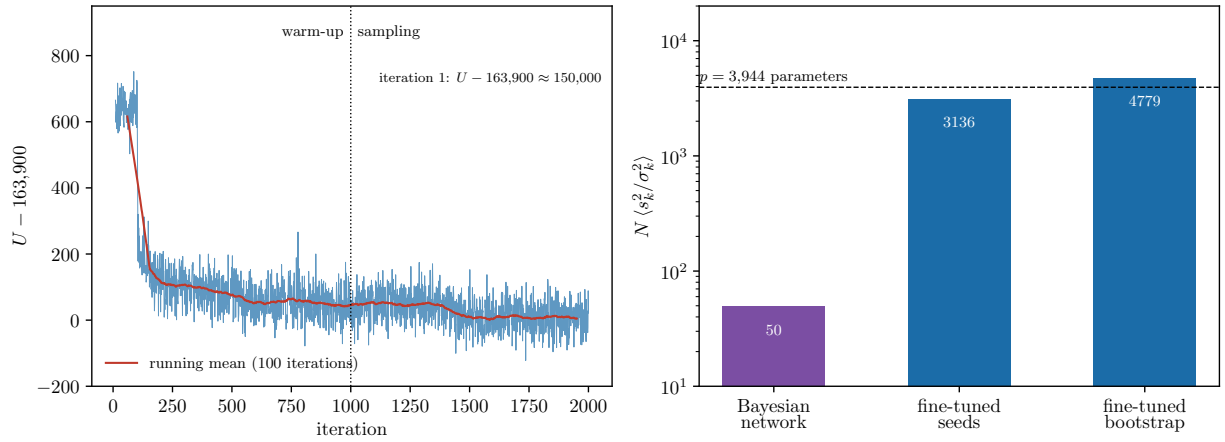


Figure 2: Left: the potential energy U of `BNN_poisson_full` from iteration 10 on (it starts 150,000 higher, at the L1 solution, and falls within ten iterations). It falls in steps, at the first mass-matrix updates, and is still drifting downwards during sampling, by about 100 over 1,800 iterations, two standard deviations of its fluctuations. Right: $N\langle s_k^2/\sigma_k^2\rangle$, an estimate of the number of well-determined parameters, eq. (1), for the Bayesian network and, for comparison, for the two fine-tuned ensembles.

3.2 The chain was still relaxing

The left panel of fig. 2 shows that the chain reached the χ^2 region within ten iterations, sat nearly still for the first 150 iterations (the step size was $\sim 10^{-7}$ until the first mass-matrix updates), and has been drifting slowly downhill ever since. A drift that persists through the whole sampling phase means the chain has not reached its stationary distribution; samples drawn while it relaxes are too tightly clustered. Note that the size of the fluctuations of U (standard deviation 45, close

to $\sqrt{p/2} = 44$ expected for a p-dimensional Gaussian) is *not* evidence of good mixing: they are dominated by the stiff directions, which mix fast, and by the Gibbs updates of the 16 precisions.

3.3 The step size

The leapfrog integrator [3] is stable only if $\varepsilon \lesssim 2/\sqrt{\lambda_{\max}}$, where $\lambda_{\max}$ is the largest curvature of U in the coordinates set by the mass matrix. If the diagonal mass matrix had equalised the curvatures, ε would be of order $p^{-1/4} \approx 0.1$ and a few tens of leapfrog steps would cross a mode. The tuned $\varepsilon \approx 8 \times 10^{-4}$ implies a remaining curvature ratio of order 10^6: strong correlations between parameters, which a diagonal mass matrix cannot remove. 800 leapfrog steps then move the soft directions by only a fraction of their width per iteration. There is a further trap: the mass matrix is estimated from the spread of the chain during warm-up, so a direction that barely moves gets a small estimated variance and moves even less in the next window.

3.4 The likelihood is too confident

At the end of the run, $U \approx 163{,}900$. The prior term $\frac{1}{2}\sum_i \tau_{g(i)}\omega_i^2$ is about $p/2 \approx 2{,}000$, the rest is the likelihood term $\frac{1}{2}\sum_k r_k^2$: $\chi^2 \approx 1.18$ per training row (with the training variances, which include the one-event term and are therefore, if anything, generous), where a correct noise model with $\gamma \ll N$ well-determined parameters should give $1 - \gamma/N \approx 1$. The training data themselves therefore say what the test spectra said: the residual variance is about 20% larger than the POWHEG noise, on average over m_{hh} (cause C).

3.5 The diagnosis in brief

Why the first attempts gave implausibly small uncertainties

1. **A, poor sampling within a mode** (the main cause): the chain's widths correspond to about 50 well-determined parameters, where a few thousand are expected; the step size shows that the diagonal mass matrix leaves curvature ratios of order 10^6; and the chain was still relaxing when sampling stopped.
2. **B, one mode**: a single chain started from one network stays near it, while 80% of the ensemble variance is between different solutions.
3. **C, an over-confident likelihood**: $\chi^2 \approx 1.18$ per training row, and the held-out model-error term of the appendix.
4. **D, the prior**: not implicated by anything we have seen, but untested; the plan tests it at no extra cost (section 4.4).

Making the chain correct would require a dense mass matrix and several much longer chains. The mixture of Laplace approximations removes A and B by construction, at a small fraction of the cost, and makes C explicit.

4 The mixture of Laplace approximations

4.1 One network: the linearised Laplace approximation

Near a network $\hat{\omega}_m$ of the ensemble, replace the network by its linearisation,

$$f(x;\omega) \approx f(x;\hat{\omega}_m) + j_m(x)^{\mathsf{T}}(\omega - \hat{\omega}_m), \qquad j_m(x) = \frac{\partial f(x;\omega)}{\partial \omega}\Big|_{\hat{\omega}_m}. \tag{2}$$

With a Gaussian likelihood of known variances σ_k^2 and the Gaussian prior of precisions τ_g, the posterior of the linearised model is Gaussian, with precision

$$H_m = \sum_{k=1}^{N} \frac{j_{m,k}\, j_{m,k}^{\mathsf{T}}}{\sigma_k^2} + \operatorname{diag}(\tau_{g(i)}) \equiv G_m + T, \tag{3}$$

where $j_{m,k} = j_m(x_k)$ and the sum runs over the training rows. G_m is the generalised Gauss–Newton matrix; it is the curvature of the χ^2 without the term proportional to the residuals, and is always positive. The prediction at any point x is then Gaussian, with mean and variance

$$\mu_m(x) = f(x;\hat{\omega}_m), \qquad v_m(x) = j_m(x)^{\mathsf{T}} H_m^{-1} j_m(x). \tag{4}$$

This is MacKay's approach to Bayesian networks [1], known today as the linearised Laplace approximation or "GLM predictive" [5, 6]. Linearising the network in the *prediction* (not only in the posterior) is what makes it work well: the predictive is that of an exactly solvable linear model, whose basis functions are the gradients of the trained network.

4.2 Twenty networks: the mixture

The posterior is approximated by a mixture of the 20 Gaussians, $p(\omega \mid \mathcal{D}) \approx \sum_m \pi_m \mathcal{N}(\omega; \hat{\omega}_m, H_m^{-1})$, with weights π_m (section 4.5) [7]. The predictive mean and variance follow from the law of total variance:

$$\bar{\mu}(x) = \sum_m \pi_m \mu_m(x), \qquad s^2(x) = \underbrace{\sum_m \pi_m v_m(x)}_{\text{within networks}} + \underbrace{\sum_m \pi_m \big(\mu_m(x) - \bar{\mu}(x)\big)^2}_{\text{between networks}}. \tag{5}$$

With equal weights the mean is the ensemble mean and the second term is the ensemble variance (with $1/M$ instead of $1/(M-1)$). The first term is what the ensemble lacks: the uncertainty of each network's parameters given the data. This is the sense in which a deep ensemble is an approximation to Bayesian model averaging [8]; the mixture of Laplace approximations completes it.

What we expect

From the identity eq. (1) and the estimate $\gamma \sim$ 3,000–4,000, the within-network term should be comparable to the between-network term, about 0.05 of the POWHEG noise, so that $s \approx \sqrt{2}$ times the ensemble spread, ten times the width the Markov chain reported. This is a prediction the computation can falsify: $\gamma_m = \operatorname{tr}(G_m H_m^{-1})$ is obtained exactly for every network.

4.3 The expansion points

The networks of the fine-tuned bootstrap ensemble minimise the Huber loss of the pulls, not χ^2 plus the prior, so they are not exactly modes of the posterior. The predictive eq. (4) is well defined at any point, and with $\delta = 2$ almost every pull lies in the quadratic region of the Huber function, so the difference is small. As an option, each network can be "polished" first: a few hundred full-batch steps of Adam on χ^2 plus the prior term $\frac{1}{2}\sum_i \tau_{g(i)}\omega_i^2$, with `finetune.py`. We compare the two on one or two networks and polish all 20 only if the widths differ appreciably.

Which variance

The likelihood uses the training variances of the fine-tune, $\sigma_k^2 = (t_k + u_k)u_k$ with $u_k = \sigma_{\text{tot}}/N_{\text{eff}}$, as in the BNN runs. The test spectra are evaluated with $t_k u_k$, as in the appendix.

4.4 The prior precisions

The prior has one precision τ_g for each of the 16 parameter tensors. Rather than copy them from a Markov chain, we determine them for each network by MacKay's evidence approximation [1], i.e. by maximising the Laplace estimate of the marginal likelihood. The fixed-point equations are

$$\tau_g = \frac{\gamma_g}{\sum_{i\in g}\hat{\omega}_{m,i}^2}, \qquad \gamma_g = n_g - \tau_g \sum_{i\in g}\left(H_m^{-1}\right)_{ii}, \tag{6}$$

where n_g is the number of parameters in tensor g and γ_g the number of them that the data determine. Only the diagonal of H_m changes with τ_g, so each iteration costs one Cholesky factorisation (about a second); a handful of iterations suffice. Holding the expansion point fixed while τ_g changes is an approximation, accurate when the prior term is small beside the χ^2, as it is here.

The prior sensitivity (cause D) comes for free: we recompute the widths with (i) the precisions of the last sample of `BNN_poisson_full`, (ii) all τ_g ten times smaller and (iii) ten times larger. If the widths at the test points move by less than their statistical uncertainty, D is excluded.

Symmetries

A ReLU network has exact flat directions: multiplying a hidden unit's incoming weights by c and its outgoing weights by $1/c$ leaves the network unchanged. In those directions G_m is zero and the width of the posterior is set by the prior alone. They do not change any prediction, so they do not enter eq. (4); but they matter for the evidence (section 4.5), and they are why H_m must include the prior to be invertible.

4.5 The weights

The Laplace estimate of the probability mass of mode m is

$$\ln Z_m \approx -\tfrac{1}{2}\chi^2(\hat{\boldsymbol{\omega}}_m) - \tfrac{1}{2}\sum_i \tau_{g(i)}\hat{\omega}_{m,i}^2 + \tfrac{1}{2}\sum_g n_g \ln\tau_g - \tfrac{1}{2}\ln\det H_m + \text{const}, \tag{7}$$

and in principle $\pi_m \propto Z_m$. In 3,944 dimensions, differences in $\ln Z_m$ of tens to hundreds are to be expected, and eq. (7) is least reliable exactly where it would make one network dominate: in the soft and flat directions, and for networks that are not exact modes. We therefore propose **equal weights**, $\pi_m = 1/20$, and report the $\ln Z_m$ as a diagnostic. If they single out a few networks

consistently (for example the ones with the lowest χ^2 on the unused training rows), we will show the evidence-weighted result alongside, and discuss it.

Is that still Bayesian?

Equal weights treat the 20 networks as an unbiased sample of the high-probability modes, the same assumption that makes the ensemble a sensible estimate of the variability of the training procedure. A choice that is closer to the data is stacking, which chooses the weights that maximise the predictive density on held-out rows [9]; the 38,076 unused training rows are available for that, and we can add it if the equal-weight result is not calibrated.

4.6 The model error

The mixture describes how well networks of this form are determined by the data. It does not describe the error of the form itself, and inflating the likelihood variance to $(1 + a)\sigma_k^2$ widens each v_m only by $(1 + a)$, far too little. As in the appendix, we add the discrepancy at the level of the prediction:

$$s^2_{\rm tot}(\boldsymbol{x}) = s^2(\boldsymbol{x}) + a(m_{hh})\,\sigma_k^2, \tag{8}$$

with $a(m_{hh})$ re-determined on the 38,076 unused training rows, now with s from eq. (5) in place of the ensemble spread, so that the test spectra are used only for validation. Since s will be larger than the ensemble spread, a will come out somewhat smaller. This term is the Bayesian analogue of a model-discrepancy function [10], treated as independent from bin to bin; the appendix will say so.

4.7 Validation

On the 1,000 test spectra, in the format of the appendix:

1. the pull s.d. with s alone (within, between, and both) and with $s_{\rm tot}$, as functions of m_{hh};
2. γ_m for every network, the identity eq. (1) checked against the sum of v_m/σ_k^2 over the training rows, and the eigenvalue spectrum of H_m;
3. **a check of the linearisation**: draw 200 parameter vectors from $\mathcal{N}(\hat{\boldsymbol{\omega}}_m, H_m^{-1})$ for one or two networks and evaluate the *full* network at the test points. If the spread agrees with $\sqrt{v_m}$, the linearisation holds over the width of the mode. (It is known that this "sampled Laplace" can be much wider than the linearised one when it fails [5]; a disagreement is a warning, not a contradiction.)
4. the width from `BNN_poisson_full` at the same points, to quantify how far the chain was from the width of its own mode.

5 Computation

- **Per-row gradients.** `torch.func.vmap(torch.func.grad(...))` gives the gradients $\boldsymbol{j}_k$ of a chunk of rows in one call: 4,096 rows give a 4,096 × 3,944 matrix, 65 MB in single precision.
- G_m**.** One matrix product per chunk, $J^{\mathsf{T}}\mathrm{diag}(1/\sigma^2)J$, accumulated in double precision: 67 chunks for 274,152 rows. G_m has 15.6 million entries (125 MB in double precision).
- **Cholesky.** $H_m = LL^{\mathsf{T}}$ in double precision; a second. The variances eq. (4) are $|L^{-1}\boldsymbol{j}|^2$: one triangular solve per chunk of test points (80,000 points).

- **Cost.** I estimate a few minutes per network on the 5090, under an hour for all 20, including the evidence iterations; the first network will tell us. On the CPU here it would take many hours, so it runs on the 5090 as a notebook.
- **Storage.** Per network: L in single precision (62 MB), the τ_g, γ_g, $\ln Z_m$, the eigenvalues, and the predictions and variances at the test spectra and the unused training rows. About 1.5 GB for 20 networks; L can be dropped if space matters.

6 What could go in the paper

Outcome	Statement
Within-network widths of a few per cent of σ_k; the linearisation check passes; pulls calibrated with s_{tot}	"Approximating the posterior by a mixture of linearised Laplace approximations about the ensemble networks, together with a model-discrepancy term, gives calibrated uncertainties. The ensemble alone omits the parameter uncertainty of each network, the Bayesian treatment supplies it." The strongest outcome.
As above, but the evidences single out a few networks	As above, with the equal-weight and evidence-weighted results side by side and a sentence on the reliability of Laplace evidences.
The linearisation check fails	Report the linearised result as an approximation and the failure honestly; a dense-mass Markov chain (the previous version of this plan) becomes the next step.
Within-network widths are as small as the chain's	The narrow width is then a property of the posterior, not of the sampler; the uncertainty is the between-network spread plus the model error, and the paper says so.

Table 2: Possible outcomes and the statement each supports.

In no case should the paper present the width of $0.005\,\sigma_k$ from the Markov chain as "the Bayesian uncertainty". A sentence on why the chain failed (section 3) is worth including: readers who have tried HMC on networks will recognise it.

7 Code

- **New module** `dihiggs/laplace.py`, commented for non-experts like the others:

```
G = gauss_newton(model, X, V, chunk_size=4096)  # G of eq. (3)
lap = Laplace(model, G, view)         # holds G, tau, Cholesky factor
lap.fit_prior(n_iter=10)              # MacKay's updates, eq. (6)
mu, v = lap.predict(X_test)           # eq. (4)
gamma = lap.gamma()                   # eq. (1)
lnZ = lap.log_evidence(chi2)          # eq. (7)
W = lap.sample(200)                   # the linearisation check
mixture_predict(laps, X_test)         # eq. (5), laps = list of 20
```

  It reuses `ParameterView` from `hmc.py` for the grouping of the parameters.

- **New notebook** `code/heft_laplace.ipynb`, with switches `ENSEMBLE` (the run folder), `MEMBERS`, `POLISH_STEPS` (0 for none), `PRIOR` (`'evidence'`, `'bnn'` or a number), `N_SAMPLES_CHECK`, `CHUNK_SIZE`, `SAVE_CHOLESKY`. It writes `runs/LAPLACE_<ensemble>/`, one folder per network plus a summary, and makes the validation tables and plots. Like the others, it can run under `nohup` from `nbconvert`.
- **A test with an exact answer** (in the notebook, or `code/laplace_toy.py`): for a model that is linear in its parameters, the Laplace posterior is exact, and `gauss_newton`, `predict` and `gamma` must reproduce the textbook formulae to rounding error.

8 Order of work

1. `laplace.py` and the exact-answer test (here, on the CPU).
2. One network on the 5090: timing, γ, widths, the linearisation check, polished versus unpolished. *Gate:* widths and γ sensible; check passes.
3. All 20 networks; the mixture; the model-error coefficient on the unused rows; the validation on the test spectra.
4. Write-up as `docs/bnn/bnn_laplace.pdf`, and a revised Bayesian subsection for the appendix.

9 Decisions for you

Decisions before I proceed

1. **The mixture of linearised Laplace approximations** about the 20 networks of the fine-tuned bootstrap ensemble (on the Mac `runs/ENS_bootstrap_huber_d2_1`), in place of further Markov chains. Agreed?
2. **Expansion points as they are**, with polishing tested on one or two networks and applied only if it matters. Agreed?
3. **Prior precisions by MacKay's evidence updates**, per network, with the BNN values and $\times 10^{\pm 1}$ as the sensitivity check. Agreed?
4. **Equal weights**, the evidences reported as a diagnostic, stacking only if needed. Agreed?
5. **The model error added to the predictive**, with $a(m_{hh})$ re-determined on the unused training rows using the mixture width. Agreed?
6. **A new module and notebook**, `dihiggs/laplace.py` and `code/heft_laplace.ipynb`, run on the 5090. Agreed?
7. **This document** replaces the previous version of `docs/bnn/bnn_plan2.pdf`, which is kept as `bnn_plan2_v1.pdf`. Acceptable?

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